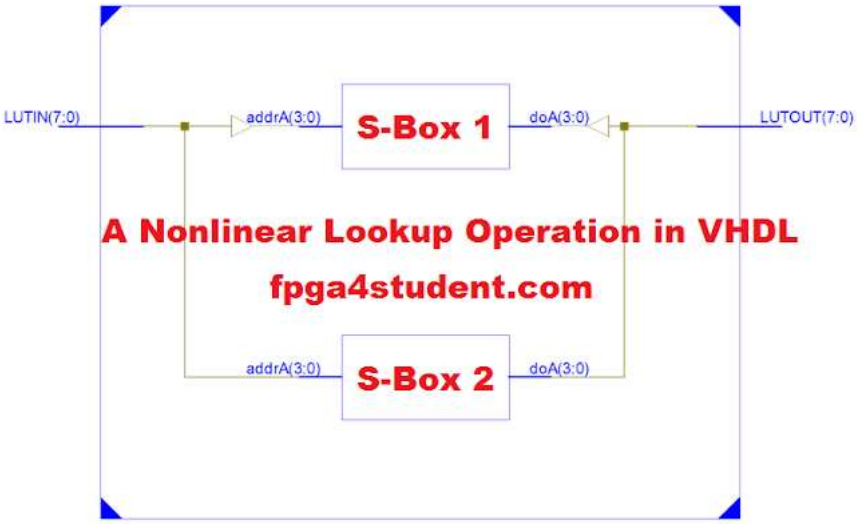
Non-linear Lookup Table Implementation in VHDL

In this VHDL project, a nonlinear lookup table which is used in hashing functions of the upcoming co-processor is implemented in VHDL.

The nonlinear operation which is used in hashing algorithm utilizes a parallel 4-bit nonlinear operation where the input nibble (4 bits) are mapped to another nonlinear 4-bit value.

The nonlinear lookup operation unit is shown in the following figure:



The details of the lookup table implementation are as follows:

S-Box 1 fpga4student.com S-Box 2

Input	Output	Input	Output
“0000”	“0001”	“0000”	“1111”
“0001”	“1011”	“0001”	“0000”
“0010”	“1001”	“0010”	“1101”
“0011”	“1100”	“0011”	“0111”
“0100”	“1101”	“0100”	“1011”
“0101”	“0110”	“0101”	“1110”
“0110”	“1111”	“0110”	“0101”
“0111”	“0011”	“0111”	“1010”
“1000”	“1110”	“1000”	“1001”
“1001”	“1000”	“1001”	“0010”
“1010”	“0111”	“1010”	“1100”
“1011”	“0100”	“1011”	“0001”
“1100”	“1010”	“1100”	“0011”
“1101”	“0010”	“1101”	“0100”
“1110”	“0101”	“1110”	“1000”
“1111”	“0000”	“1111”	“0110”

VHDL code for the nonlinear lookup table implementation:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL implementation of Lookup Table
---------------------------------------------------
-- Non-linear Lookup Table Implementation in VHDL--
---------------------------------------------------
entity non_linear_lookup is
port (  LUTIN: in std_logic_vector(7 downto 0);
   LUTOUT: out std_logic_vector(7 downto 0)
 );
end non_linear_lookup;
```

```
architecture Behavioral of non_linear_lookup is
signal MSN_in,LSN_in,MSN_out,LSN_out: std_logic_vector(3 downto 0);
begin
MSN_in <= LUTIN(7 downto 4);
LSN_in <= LUTIN(3 downto 0);
SBOX_1: process(MSN_in) begin
case(MSN_in) is
 when "0000" => MSN_out <= "0001";
 when "0001" => MSN_out <= "1011";
 when "0010" => MSN_out <= "1001";
 when "0011" => MSN_out <= "1100";
 when "0100" => MSN_out <= "1101";
 when "0101" => MSN_out <= "0110";
 when "0110" => MSN_out <= "1111";
 when "0111" => MSN_out <= "0011";
 when "1000" => MSN_out <= "1110";
 when "1001" => MSN_out <= "1000";
 when "1010" => MSN_out <= "0111";
 when "1011" => MSN_out <= "0100";
 when "1100" => MSN_out <= "1010";
 when "1101" => MSN_out <= "0010";
 when "1110" => MSN_out <= "0101";
 when "1111" => MSN_out <= "0000";
 when others => MSN_out <= "0000";
end case;
end process;
SBOX_2: process(LSN_in) begin
case(LSN_in) is
 when "0000" => LSN_out <= "1111";
 when "0001" => LSN_out <= "0000";
 when "0010" => LSN_out <= "1101";
 when "0011" => LSN_out <= "0111";
 when "0100" => LSN_out <= "1011";
 when "0101" => LSN_out <= "1110";
 when "0110" => LSN_out <= "0101";
 when "0111" => LSN_out <= "1010";
 when "1000" => LSN_out <= "1001";
 when "1001" => LSN_out <= "0010";
 when "1010" => LSN_out <= "1100";
 when "1011" => LSN_out <= "0001";
 when "1100" => LSN_out <= "0011";
 when "1101" => LSN_out <= "0100";
 when "1110" => LSN_out <= "1000";
 when "1111" => LSN_out <= "0110";
 when others => LSN_out <= "0000";
end case;
end process;
LUTOUT <= MSN_out & LSN_out;
end Behavioral;
```

VHDL Testbench code for Nonlinear Lookup Table implementation:

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
USE ieee.numeric_std.all;
-- fpga4student.com: FPGA projects, Verilog projects, VHDL projects
-- VHDL project: VHDL implementation of Lookup Table
-- Testbench VHDL code for Lookup Table Implementation
ENTITY tb_lookuptable IS
END tb_lookuptable;

ARCHITECTURE behavior OF tb_lookuptable IS

    -- Component Declaration for Lookup Table Implementation in VHDL
```

```
    COMPONENT non_linear_lookup
    PORT(
         LUTIN : IN  std_logic_vector(7 downto 0);
         LUTOUT : OUT  std_logic_vector(7 downto 0)
        );
    END COMPONENT;


   --Inputs
   signal LUTIN : std_logic_vector(7 downto 0) := (others => '0');

  --Outputs
   signal LUTOUT : std_logic_vector(7 downto 0);
 signal i: integer;

BEGIN

 -- Instantiate the Lookup Table Implementation
   uut: non_linear_lookup PORT MAP (
          LUTIN => LUTIN,
          LUTOUT => LUTOUT
        );

   stim_proc: process
   begin
  LUTIN <= x"00";
  -- initialize 4-bit input data in VHDL testbench
      for i in 0 to 15 loop
   LUTIN <= std_logic_vector(to_unsigned(i, 8));
  wait for 100 ns;
  end loop;
      wait;
   end process;

END;
```

Simulation waveform for the nonlinear lookup table in VHDL:

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